

Anthony Liang

[aliang8.github.io](https://github.com/aliang8) ♦ anthony.liang@usc.edu
[linkedin.com/in/ant-liang](https://www.linkedin.com/in/ant-liang)

RESEARCH INTERESTS

My research focuses on scalable robot learning, with an emphasis on data-efficient human feedback (gaze, demonstrations, preferences, interventions) and reward foundation models that provide generalizable reward functions across tasks, embodiments, and environments. I also work on core components for embodied intelligence, including vision-language-action models for low-level control, trace models for structured planning, and world models for scene understanding and predictive reasoning. In parallel, I am investigating RL post-training for LLM-based web agents to improve reliability in sequential UI tasks and long-horizon interactions.

EDUCATION

University of Southern California , Los Angeles, CA <i>Ph.D.</i> in Computer Science (Co-advised by Erdem Biyik and Stephen Tu)	Aug 2021 - Present <i>GPA: 3.9/4.0</i>
University of Michigan Rackham Graduate School , Ann Arbor, MI <i>M.S.</i> in Robotics (Advisor: Honglak Lee)	Aug 2017 - 2021 <i>GPA: 4.0/4.0</i>
University of Michigan , Ann Arbor, MI <i>B.S.E.</i> (Advisor: Honglak Lee)	Aug 2017 - 2021 <i>GPA: 3.65/4.0</i>

RESEARCH EXPERIENCE

LiRA Lab and Statistical Learning Lab , USC <i>Ph.D. Student, PI: Erdem Biyik, Stephen Tu</i>	Aug 2021 - Present
· Robot learning with multimodal human feedback, sample-efficient RL, learning from unlabelled videos	
Intelligent Robot Lab , Carnegie Mellon <i>Visiting Researcher</i> with Changliu Liu	May 2020 - May 2021
· Hierarchical RL for safe control of autonomous vehicles in dynamic environments	
Deep Learning Lab , University of Michigan <i>Research Intern</i> with Honglak Lee	Jan 2019 - May 2021
· Sample-efficient RL for embodied task learning	

PROFESSIONAL EXPERIENCE

Google DeepMind <i>Research Intern</i> with Kalpesh Krishna, Jacob Eisenstein	May 2025 - Sept 2025 <i>NYC/Seattle</i>
· RL post-training for reasoning models	
Google Research <i>Research Intern</i> with Chih-Wei Hsu, Yinlam Chow, Guy Tennenholtz, Craig Boutilier	May 2023 - Feb 2024 <i>Remote NYC</i>
· Bayesian RL for Markov Decision Processes with gradually changing latent dynamics	
· Data augmentation at critical states for robot imitation learning with Stephen Tu	
· Generative modeling for online RL policies	
Meta AI - Multimodal Learning Team <i>Research Intern</i> with Paul Crook and Andrea Madatto	May 2022 - Aug 2022 <i>Redmond, WA</i>
· Fine-tuning large language models for task-oriented dialogue generation	
Amazon Science <i>Applied Science Intern</i> with Thiago Mosquero	May 2021 - Aug 2021 <i>Seattle, WA</i>

- Collaborative filtering for recommending new brands and products to consumers

Invisible.ai
AI Research Intern

May 2020 - Aug 2020
Remote

- Improving computer vision models for real-time object detection and tracking for industrial processes

Google Ads
Software Engineering Intern

May 2019 - Aug 2019
Mountain View, CA

Luminar Technologies
Software Engineering Intern

May 2018 - Aug 2018
Palo Alto, CA

PUBLICATIONS

-
- [C9] **Anthony Liang***, Yigit Korkmaz*, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sid Kaushik, Aditya Shah, Alex Huang, Luke Zettlemoyer, Dieter Fox, Yu Xiang, Anqi Li, Andreea Bobu, Abhishek Gupta, Stephen Tu[†], Erdem Biyik[†], Jesse Zhang[†]. "Robometer: Scaling General-Purpose Robotic Reward Models via Trajectory Comparisons". *Robotics: Science and Systems (RSS) 2026*
 - [C8] **Anthony Liang**, Jonathan Berant, Adam Fisch, Abhimanyu Goyal, Kalpesh Krishna[†], Jacob Eisenstein[†]. "Plantain: Plan-Answer Interleaved Reasoning". *International Conference on Machine Learning (ICML) 2026 (Spotlight)*
 - [C7] Mihir Rao*, **Anthony Liang***, Abrar Anwar*, Rudy Corona, Giscard Biamby, Dantong Liu, Roei Herzig, Trevor Darrell. "In-Context Learning Trace Model for Robot Learning" *In preparation*
 - [C6] Matthew Hong*, **Anthony Liang***, Kevin Kim, Harshitha Rajaprakash, Jesse Thomason[†], Erdem Biyik[†], Jesse Zhang[†]. "HAND Me the Data: Fast Robot Adaptation via Hand Path Retrieval" *International Conference on Robot Automation (ICRA) 2026*
 - [C5] Jaiv Doshi*, **Anthony Liang***, Yigit Korkmaz, Erdem Biyik. "RHODES: Reducing Human Oversight via Disagreement and Exploration for Safe Reinforcement Learning" *Preprint*
 - [C4] **Anthony Liang***, Pavel Czempin*, Matthew Hong, Yutai Zhou, Erdem Biyik, Stephen Tu. "CLAM: Continuous Latent Action Models for Robot Learning from Unlabeled Demonstrations" *In submission*
 - [C3] **Anthony Liang**, Chih-wei Hsu, Yinlam Chow, Guy Tennenholtz, Erdem Biyik, Craig Boutilier. "DynaMITE-RL: A Dynamics Model for Improved Temporal Meta Reinforcement Learning", *International Conference on Machine Learning (ICML) AutoRL Workshop 2024, Conference on Neural Information Processing Systems (NeurIPS) 2024*
 - [C2] **Anthony Liang**, Jesse Thomason, Erdem Biyik. "ViSaRL: Visual Reinforcement Learning Guided By Human Saliency", *Spotlight talk at ICRA Pretraining for Robotics (PT4R) Workshop, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024*
 - [C1] Wilka Carvalho, **Anthony Liang**, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. "Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in a First-person Simulated 3D Environment", *International Joint Conferences on Artificial Intelligence (IJCAI) 2021*
 - [T5] **Anthony Liang**, Yigit Korkmaz, Jiahui Zhang, Jesse Zhang, Abrar Anwar, Sidhant Kaushik, Yufei Wang, Yu Xiang, David Held, Dieter Fox, Abhishek Gupta, Stephen Tu, Erdem Biyik. "SPUR: Scaling Reward Learning from Human Demonstrations". *CoRL Eval&Deploy Workshop 2025*
 - [T4] **Anthony Liang**, Pavel Czempin, Yutai Zhou, Stephen Tu, Erdem Biyik. "In-Context Generalization to New Tasks From Unlabeled Observation Data", *ICML In-Context Learning Workshop 2024*
 - [T3] Ishika Singh, **Anthony Liang**, Mohit Shridhar, Jesse Thomason. "Self-Supervised 3D Representation Learning for Robotics", *ICRA Pretraining for Robotics (PT4R) 2023*
 - [T2] **Anthony Liang**, Ishika Singh, Karl Pertsch, Jesse Thomason. "Transformer Adapters for Robot Learning", *CoRL Workshop on Pretraining for Robot Learning 2022*

- [T1] Wilka Carvalho, **Anthony Liang**, Kimin Lee, Sungryull Sohn, Richard L. Lewis, Satinder Singh, Honglak Lee. "ROMA: A Relational, Object-Model Learning Agent for Sample-Efficient Reinforcement Learning", *ICML Workshop on Object-Oriented Learning 2020*

TEACHING

University of Southern California

CSCI 445: Foundation of Robotics	Fall 2026
CSCI 699: Robot Learning	Fall 2024
CSCI 499: Natural Language for Interactive AI	Fall 2022

University of Michigan, Ann Arbor

EECS 442: Computer Vision	Winter 2021
EECS 498: Algorithmic Robotics	Fall 2020
EECS 504: Graduate Computer Vision	Winter 2020
EECS 280: Introduction to Programming and Data Structures	Fall 2018 - Fall 2019
Summer STEM Institute Research Mentor	Summer 2021

SERVICES

Reviewer:

- ICML, ICLR, ICRA, NeurIPS, HRI, RA-L, AAAI, RO-MAN, CVPR

Workshop Organizer:

- Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting at RSS 2025

Mentorship and Outreach:

- USC UGrad Mentoring Program
- CURVE: Mentored USC undergrads through fellowship program.

STUDENT MENTORSHIP

USC Masters Students

- | | |
|--------------------------|-------------------------------------|
| • Matthew Hong | Learning from unlabelled data, RLHF |
| • Dhanush Kumar Penmetsa | Gaze for robot teleoperation |
| • Ziyi Liu | Preference-based RL |

USC Undergraduate Students

- | | |
|------------------------------------|---|
| • Yixi Quan (Undergrad, USC) | Pretrained Video-LLMs for Robot Learning |
| • Jaiv Doshi (Undergrad, USC) | Human-intervention reinforcement learning |
| • Junu Song (USC CURVE Fellowship) | Real-world robot navigation |

Fellowship Programs and External Collaborators

- | | |
|---|-------------------------------|
| • Mihir Rao (Undergrad, Berkeley) | Trace VLAs |
| • Sankalp (Sunny) Agrawal (Undergrad, USC SURE Program) | Meta-RL with task descriptors |
| • Shreya Ramanujam (Undergrad, IIT) | Gaze for robot teleoperation |

INVITED TALKS

"Scaling Reward Models via Preference Comparisons"

2026

- Apple Robotics Reading Group (Host: Soheil Habibiian)

HONORS AND AWARDS

- | | |
|--|------|
| • NSF Graduate Research Fellowship Honorable Mention | 2020 |
|--|------|